🔑 **Answer Key — Teacher Copy · Lesson 3-9**

MSTAR PRACTICE WORKSHEET

Lesson 3-9: Equivalent Ratios

Grade 6 Mathematics · Standard 6.AT.3

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

Which ratio is equivalent to 2:5?

A. 2:10

B. 3:6

C. 4:10

D. 5:2

Multiply both parts by 2: 2×2 = 4 and 5×2 = 10, so 4:10 is equivalent to 2:5.

Question 2 SELECTED RESPONSE — CORRECT: A

A recipe uses 4 eggs for every 6 cups of flour. Which ratio is equivalent?

A. 2:3

B. 4:3

C. 6:4

D. 8:10

Divide both by 2: 4÷2 = 2 and 6÷2 = 3. The ratio 2:3 is equivalent to 4:6.

Question 3 SELECTED RESPONSE — CORRECT: B

Which ratio is NOT equivalent to 6:9?

A 2:3

B 3:5

C 12:18

D 18:27

6:9 simplifies to 2:3. Check each: 2:3 ✓, 12:18 = 2:3 ✓, 18:27 = 2:3 ✓. But 3:5 does not simplify to 2:3.

Question 4 SELECTED RESPONSE — CORRECT: B

What is the simplest form of the ratio 15:25?

A 1:5

B 3:5

C 5:3

D 15:25

The GCF of 15 and 25 is 5. Divide both by 5: 15÷5 = 3 and 25÷5 = 5. The simplest form is 3:5.

Question 5 SELECTED RESPONSE — CORRECT: C

Marcus earns $45 for every 3 hours of work. How much does he earn in 8 hours?

A $90

B $105

C $120

D $135

Unit rate: $45 ÷ 3 = $15/hour. In 8 hours: $15 × 8 = $120.

Question 6 SELECTED RESPONSE — CORRECT: D

Chef Montoya's dressing uses 3 spoons of oil to 4 spoons of vinegar (3:4). Which recipe card shows an EQUIVALENT ratio?

A 3:8

B 4:3

C 6:7

D 9:12

Multiply both parts of 3:4 by 3: 3 x 3 = 9 and 4 x 3 = 12, so 9:12 is equivalent to 3:4.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: B

Solve the proportion 4/10 = x/25 for x.

A 8

B 10

C 19

D 62.5

The scale factor from 10 to 25 is 2.5, so x = 4 × 2.5 = 10. Cross products confirm it: 4 × 25 = 100 and 10 × 10 = 100.

- **A:** That doubles the 4. The denominator went from 10 to 25, a factor of 2.5 rather than 2.
- **C:** That adds the 15 that was added to the denominator. Proportions scale by multiplying.
- **D:** That scales the 25 instead of the 4. Apply the factor 2.5 to the numerator.

Part B — correct answer: B

How do cross products show that two ratios are equivalent?

A If a + d equals b + c, the ratios are equivalent.

B If a/b = c/d, then a × d equals b × c.

C Cross products work only when all four numbers are whole.

D Cross products compare the two numerators only.

Multiplying both sides of a/b = c/d by b·d gives a·d = b·c, so equal cross products mean the ratios are equivalent.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, D

Select ALL pairs of ratios that are equivalent:

- **A** 2/3 and 8/12
- **B** 5/8 and 10/16
- **C** 3/4 and 4/5
- **D** 6/9 and 2/3
- **E** 7/10 and 14/21

A (2×12 = 24 = 3×8), B (5×16 = 80 = 8×10), and D (6×3 = 18 = 9×2) are equivalent. C fails: 3×5 = 15 but 4×4 = 16. E fails: 7×21 = 147 but 10×14 = 140.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A store sells 3 pens for $2. Another store sells 9 pens for $6. Are these equivalent ratios? If so, prove it using at least two different methods (simplifying, scaling, or cross-multiplying).

Model answer: Yes, they are equivalent. Method 1 (simplify): 9:6 ÷ 3 = 3:2, which matches 3:2. Method 2 (cross-multiply): 3 × 6 = 18 and 2 × 9 = 18. Since both cross products are equal, the ratios are equivalent.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.